ResearchPilot: Autonomous Deep Scientific Literature Agents via Gemma 4

Abstract
We present ResearchPilot AI, an autonomous research assistant powered by Gemma 4.
Scientific literature is growing at over 3 million papers annually. Current language models often hallucinate experimental results, fabricate metrics, or misquote baseline citations.
ResearchPilot addresses this by pairing strict lexical boundary extraction with grounded Gemma 4 reasoning, multimodal figure inspection, and bounded agentic tool execution.

1. Introduction
Researchers spend up to 40% of their time scanning scientific papers, parsing dense tables, and extracting methodological parameters. To accelerate research comprehension while guaranteeing factual fidelity, we formalize a grounded research-assistant architecture.

2. Methodology and Architecture

ResearchPilot consists of four modular components:
1. Local PyMuPDF text and page extraction with strict [PAGE N] indexing.
2. Google Gen AI SDK client targeting Gemma 4 (gemma-4-26b-a4b-it).
3. High-resolution page renderer for visual diagram inspection.
4. Bounded Agentic Tool Registry containing 5 read-only inspection tools:
   - get_paper_metadata
   - get_page_text
   - search_paper_text
   - get_paper_statistics
   - render_pdf_page

3. Experimental Setup & Datasets

We evaluate ResearchPilot across 500 peer-reviewed papers from arXiv and PubMed.
Evaluation benchmarks measure extraction fidelity, citation precision, and grounded answer accuracy.
We test on the ArXiv-SciBench dataset containing 12,000 question-answer pairs.

4. Quantitative Results & Evaluation

ResearchPilot demonstrates significant gains in factual adherence:
- Citation Precision: 98.4% compared to 71.2% for ungrounded baselines.
- Hallucination Rate: Reduced to 1.1% on quantitative benchmark metrics.
- User Comprehension Speed: 3.4x faster literature review completion times.

Table 1: Benchmark Comparison

Model / System	Citation Acc (%)	Hallucination (%)
Ungrounded Baseline	71.2%	18.4%
RAG Naive Chunking	84.6%	6.8%
ResearchPilot (Gemma 4)	98.4%	1.1%

5. Limitations

The primary limitation acknowledged by the authors is that scanned image-only PDFs require external optical character recognition (OCR). In addition, papers exceeding 100 pages require iterative chunking due to context length bounds.

6. Conclusion

ResearchPilot AI establishes a new benchmark for evidence-grounded scientific exploration.